Name _____ Period _____ Date _____

2-7 Lesson Quiz: Linear-Quadratic Systems

short form

This form has Savvas items 1, 2 and 5. Items 3 and 4 are on the full Savvas quiz only. Item 5 needs a graphing calculator (or Desmos).

1. Determine the number of real solutions of the system $y = x^2 - 3$, $y = -2x + 4$.

Ⓐ 2

Ⓑ 1

Ⓒ 3

Ⓓ 0

Show how you know (a sketch or your algebra):

_____

_____

_____

_____

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2. Use substitution to solve the system $y = -\frac{1}{2}x^2$, $y = x - 4$.

Ⓐ -4 and 2

Ⓑ $(4, -8)$ and $(-2, -2)$

Ⓒ $(-4, -8)$ and $(2, -2)$

Ⓓ $(0, 0)$ and $(0, -4)$

Show your substitution:

_____

_____

_____

_____

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5. Solve the equation $2x^2 + 5x - 8 = \frac{5}{2}x + 20$ by writing a linear-quadratic system and solving using the intersection feature of a graphing calculator. Round to the nearest tenth.

The system I graphed: $y =$ _____ $y =$ _____

The graphing calculator shows the curve and line intersect at $x \approx$ _____ and $x \approx$ _____.

Scoring: Questions 1 and 5 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

2-7 Lesson Quiz, short form: answer key

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1. A, 2. $x^2 - 3 = -2x + 4$, so $x^2 + 2x - 7 = 0$; the discriminant is $2^2 - 4(1)(-7) = 32 > 0$, so two real solutions ($x = -1 \pm 2\sqrt{2}$, about 1.83 and -3.83). A sketch shows the line crossing the parabola (vertex $(0, -3)$) twice. DOK 2, Realize skills A91, A54; IXL VQU (Example 1). **10 points:** 6 for A, 4 for a reason (a sketch with two crossings, or the discriminant).

2. C, $(-4, -8)$ and $(2, -2)$. $-\frac{1}{2}x^2 = x - 4$, so $x^2 + 2x - 8 = 0$, $(x + 4)(x - 2) = 0$, $x = -4$ or $x = 2$; then $y = x - 4$ gives -8 and -2. Distractor A is the x-values without their y-values; B swaps the signs of x. DOK 2, Realize skills A91, A54; IXL HVZ (the Example 1 check; the packet's Exit question). **Bonus 5:** 5 for C with the substitution shown; 3 for C alone; 2 for A with correct work (the x-values found, not paired).

5. $x \approx 3.2$ and $x \approx -4.4$. Graph $y = 2x^2 + 5x - 8$ and $y = \frac{5}{2}x + 20$; the intersections are about $(3.17, 27.92)$ and $(-4.42, 8.96)$. Check: $2x^2 + \frac{5}{2}x - 28 = 0$ gives $x = \frac{-2.5 \pm \sqrt{230.25}}{4}$, about 3.168 and -4.418. DOK 2, Realize skills A91, A54; IXL VQU (Example 5). **10 points:** 2 for the system, 4 for each x-value to the nearest tenth (2 each if not rounded to tenths).

Items left off this form (full Savvas quiz only; available for a retake or Form B): Savvas item 3, Nate tosses a ball up a hill, path $y = -\frac{1}{4}x^2 + \frac{33}{5}x$, hill $y = \frac{1}{5}x$: how far does the ball travel horizontally (answer 25.6 ft; DOK 1, A91, A54; taught by Example 3, cut in the fast packet); Savvas item 4, solve the system of inequalities $y \geq 2x^2 + 2x - 3$, $y \leq \frac{1}{3}x + 6$ using shading (answer: both boundaries solid, shade above the parabola and below the line where they overlap; DOK 2, A91, A54; taught by Example 4, cut).

Topic 2 test: item 1 is the same task as Topic Assessment question 16, and item 5 the same task as question 17 (to the nearest hundredth there).

IXL replacement: SmartScore 70 or above on VQU and HVZ replaces this quiz with 20/20.